

| Assessment | Test | Domain                            | Skill                                                | Difficulty                               |
|------------|------|-----------------------------------|------------------------------------------------------|------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Ratios, rates, proportional relationships, and units | Hard <div></div> <div></div> <div></div> |

Question

For an electric field passing through a flat surface perpendicular to it, the electric flux of the electric field through the surface is the product of the electric field's strength and the area of the surface. A certain flat surface consists of two adjacent squares, where the side length, in meters, of the larger square is **3** times the side length, in meters, of the smaller square. An electric field with strength **29.00** volts per meter passes uniformly through this surface, which is perpendicular to the electric field. If the total electric flux of the electric field through this surface is **4,640 volts · meters**, what is the electric flux, in **volts · meters**, of the electric field through the larger square?

Correct Answer: 4176

Rationale

The correct answer is **4,176**. It's given that the side length of the larger square is **3** times the side length of the smaller square. This means that the area of the larger square is **3<sup>2</sup>**, or **9**, times the area of the smaller square. If the area of the smaller square is represented by ***x***, then the area of the larger square can be represented by **9*x***. Therefore, the flat surface of the two adjacent squares has a total area of ***x* + 9*x***, or **10*x***. It's given that an electric field with strength **29.00** volts per meter passes uniformly through this surface and the total electric flux of the electric field through this surface is **4,640 volts · meters**. Since it's given that the electric flux is the product of the electric field's strength and the area of the surface, the equation **29.00(10*x*) = 4,640**, or **290*x* = 4,640**, can be used to represent this situation. Dividing each side of this equation by **290** yields ***x* = 16**. Substituting **16** for ***x*** in the expression for the area of the larger square, **9*x***, yields **9(16)**, or **144**, square meters. Since the area of the larger square is **144** square meters, the electric flux, in **volts · meters**, of the electric field through the larger square can be determined by multiplying the area of the larger square by the strength of the electric field. Thus, the electric flux is **(144 square meters)( $\frac{29.00 \text{ volts}}{\text{meter}}$ )**, or **4,176 volts · meters**.